THE MASTERMIND COACHING CLASSES

Revision Notes

Class 7 · Social Science

Exploring Society: India and Beyond — Chapters 1 to 6 · CBSE / RBSE Syllabus

HOW TO USE THESE NOTES

Each chapter follows the textbook's own order of sections, so you can read straight through. Bold words are keywords you must be able to define. Tables compare and classify; shaded boxes hold definitions, formulas and key facts. The navy Key Takeaways box at the end of every chapter is your last-two-minutes revision.

CH.	CHAPTER TITLE	CORE THEME
1	Geographical Diversity of India	Himalayas, plains, desert, plateau, coasts, islands
2	Understanding the Weather	Elements of weather, instruments, forecasting
3	Climates of India	Climate factors, monsoons, disasters, climate change
4	How the Land Becomes Sacred	Sacred places, pilgrimage, sacred ecology
5	New Beginnings: Cities and States	Second Urbanisation, mahajanapadas, early republics, varna-jati
6	The Rise of Empires	Empires, Magadha, Alexander, Mauryas, Kautilya, Ashoka

The Mastermind Coaching Classes — Road No. 5, VKIA, Jaipur — 9887258679

CHAPTER 01 · GEOGRAPHY

Geographical Diversity of India

India is large, old and varied. Walking across its map from the snowy Himalayas to the coral islands shows how mountains, rivers, deserts, plateaus and coasts have shaped the way people live, farm, trade, build and worship. This chapter is a bird's-eye tour of those landforms.

Big Questions

- What are the key geographical features of India?
- How does India's geographical diversity affect our lives?

1.1 India on the Map

- India is the **seventh-largest country** in the world and lies in **Asia**.
- With Pakistan, Bangladesh, Nepal, Bhutan, Sri Lanka and Myanmar it forms the **Indian Subcontinent** (called a *sub*continent because it is a part of the continent of Asia).
- Natural boundaries: the **Himalayas** in the north, the **Thar Desert and Arabian Sea** in the west, the **Indian Ocean** in the south and the **Bay of Bengal** in the east. These separate India from the rest of Asia and have shaped its climate, culture and history.
- In 1984 **Rakesh Sharma**, India's first astronaut, told Indira Gandhi that India looks *'Sare jahan se achha'* from space.

For study, the book groups India into **five regions**: the great mountain zone, the plains of the Ganga and the Indus, the desert, the southern peninsula, and the islands. Always read the **legend (colour key)** of the physical map: each colour shows a height band.

1.2 The Himalayas

- A wall-like range about **2,500 km** long, spread over **six countries**: India, Nepal, Bhutan, China, Pakistan and Afghanistan.
- Many peaks cross **8,000 m** and are called the **'Eight Thousanders'**; the highest mountain in the world, **Mount Everest**, is here.
- Called the **'Water Tower of Asia'**: summer snow-melt feeds the **Ganga, Indus and Brahmaputra** and their tributaries, which supply water for drinking, farming and industry to hundreds of millions.
- Mountains are held **sacred**; temples and monasteries attract pilgrims and seekers. The Bhagirathi (a main stream of the Ganga) begins at **Gaumukh**, the snout of the **Gangotri Glacier** in Uttarakhand.
- The name comes from Sanskrit **hima** (snow) + **alaya** (abode): *abode of snow*.

▸ How the Himalayas were formed

- India was once part of the huge landmass **Gondwana**, next to Africa. It broke away and drifted **north**.
- About **50 million years ago** it collided with **Eurasia**; the land between crumpled and rose, like a carpet wrinkling when pushed. This is **folding**.
- India still pushes into Asia at about **5 cm a year**, so the Himalayas rise by roughly **5 mm a year** (about 5 m in a thousand years).

▸ Three ranges of the Himalayas

Range	Also called	Features	Examples
Himadri	Greater Himalayas	Highest and most rugged; snow-covered all year; few settlements	Mount Everest, Kanchenjunga
Himachal	Lower Himalayas	South of Himadri; moderate climate, rich biodiversity, people live here	Hill stations: Nainital, Mussoorie, Shimla, Darjeeling
Shivalik	Outer Himalayas	Lowest and outermost; rolling hills, dense forests, wildlife; transition to the Gangetic plains	Foothills along the plains

- **Kath-kuni / dhajji-dewari** houses of the western Himalaya use local stone and timber: they keep warm and resist mild earthquakes.
- **Great Himalayan National Park** (Himachal Pradesh) is a **UNESCO World Heritage Site** with rich flora and fauna, protected by the government together with village communities. Wildlife includes the Himalayan monal, snow leopard and ibex.
- **Yaks** are reared for milk, meat, wool and dung and used for transport.

▸ Ladakh: the cold desert

- Winter temperatures fall below **-30 °C**; rainfall is very low; terrain is rocky with deep valleys and lakes such as **Pangong Tso** (*tso* = lake; salty because of dissolved minerals).
- Called **'moonland'** for its moon-like landscape. The rocks were once under an ocean and are mostly sand and clay; **wind and rain eroded** them into strange shapes.
- Wildlife: snow leopard, ibex, Tibetan antelope. Culture: ancient monasteries and festivals such as **Losar and Hemis**.

1.3 The Gangetic Plains

- Flat, **fertile** land south of the Himalayas, watered by the **Ganga, Indus and Brahmaputra** systems. Rivers deposit **minerals (silt)** that enrich the soil.
- Ideal for **agriculture** (multi-cropping) and home to a **large share of India's population**; rivers also give hydroelectric power.
- Flat land made it easy to build **roads and railways**; rivers have been routes of **travel and trade** for thousands of years.
- Most rivers carry goddess names (Ganga, Yamuna, Kaveri); **Brahmaputra** means 'son of Brahma'. It swells in summer because of snow-melt in the Himalayas.
- Wildlife: Bengal tiger (revived by **Project Tiger**), gharial (endangered, protected by law), grey langur, peacock (**national bird**).

1.4 The Thar Desert and the Aravallis

▸ Thar (Great Indian) Desert

- Spreads over **Rajasthan, Gujarat, Punjab and Haryana**; vast and arid, with **hot days and cold nights**.
- **Sand dunes** are shaped by wind and can rise to **150 m**. The desert is a natural barrier that discourages movement of people and animals.
- Adaptation: sand is used to scour utensils to save water; rinse water is reused; Rajasthan has **taanka / kunds** (rainwater-storage structures).

- **Jaisalmer**, the 'Golden City', has a fort that is a **UNESCO World Heritage Site**; the **Pushkar Mela** (edge of the Thar) is famous for camels.

▸ Aravalli Hills

- Among the **oldest mountains in the world** (about **2.5 billion years**). Highest peak **Mount Abu** (over 1,700 m); most hills are 300-900 m.
- Spread through **Delhi, Haryana, Rajasthan and Gujarat**; they **stop the Thar from spreading east**.
- Rich in **marble, granite, zinc and copper**. The ancient mines at **Zawar** show that Indians were first to master **zinc extraction** over eight centuries ago.
- Forts: **Chittorgarh, Kumbhalgarh, Ranthambore**; hills gave natural defence.

1.5 The Peninsular Plateau

- A **plateau** is a raised land with a more or less flat top and steep sides. The triangular plateau of central-south India is surrounded by water on three sides (**peninsula**: Arabian Sea, Bay of Bengal, Indian Ocean), so it is the **Peninsular Plateau**, a very old landform.
- Bordered by the **Western Ghats** (taller, wall-like along the west coast, waterfalls in monsoon; northern part called **Sahyadri**; a **UNESCO World Heritage Site**) and the **Eastern Ghats** (lower, broken hills on the east).
- Between them lies the **Deccan Plateau**. The plateau tilts slightly **east**.

Direction of flow	Rivers	Drain into
East-flowing	Godavari, Krishna, Kaveri, Mahanadi	Bay of Bengal
West-flowing	Narmada, Tapti	Arabian Sea

- Rich in **minerals (coal), forests and fertile soil**; waterfalls give **hydroelectric power** and irrigation. Coal is a fossil fuel and adds to global warming.
- Dense forests shelter tribal communities such as the **Santhal, Gond, Baiga, Bhil and Korku**, with their own languages and nature-linked traditions. Tribal communities are spread across Jharkhand, West Bengal, Odisha, Assam, Madhya Pradesh, Chhattisgarh, Maharashtra, Telangana and Gujarat.
- Western Ghats biodiversity: lion-tailed macaque, king cobra, insect-eating plants. **Mumbai** on the west coast is India's financial centre.

1.6 India's Coastlines

India's coastline is **over 7,500 km** long, with beaches, cliffs, forests and coral reefs.

Feature	West Coast	East Coast
Stretch	Gujarat to Kerala through Maharashtra, Goa, Karnataka	Ganga delta to Kanyakumari
Lies between	Western Ghats and Arabian Sea	Eastern Ghats and Bay of Bengal
Rivers	Short, swift rivers from the Western Ghats; form **estuaries** (Narmada and Tapti are the largest)	Rivers form fertile **deltas** (Mahanadi, Godavari, Krishna, Kaveri)
Coast shape	Narrow; coves, creeks, estuaries; hills close to the sea	Wide plains
Water bodies / use	Many ports and cities, centres of trade for thousands of years	**Chilika** and **Pulicat** lakes (Pulicat is a lagoon)

DEFINITION

Delta: a triangular or fan-shaped landform formed at a river mouth when sediments settle in a larger water body. **Lagoon**: water separated from the sea by a natural barrier. **Estuary**: the wide river mouth where river meets the sea without forming a delta.

1.7 The Indian Islands

Island group	Sea	Nature	Key facts
Lakshadweep	Arabian Sea, near the Malabar coast of Kerala	**Coral** islands; an archipelago of **36** islands	Not all inhabited; gives India a large marine area for fishing and resources
Andaman and Nicobar	Bay of Bengal	**500+ volcanic** islands in two groups	Strategic outposts watching the ocean; **Cellular Jail** held freedom fighters; **Barren Island** is India's only active volcano

1.8 The Sundarbans and the North-East Hills

- **Sundarbans**: delta of the Ganga and Brahmaputra where river, sea and land meet; about half is in India, half in Bangladesh; **mangroves**; home of the **Royal Bengal Tiger**; a **UNESCO site**.
- **North-East hills (Meghalaya Plateau: Garo, Khasi, Jaintia)**: lush, one of the world's **highest-rainfall** regions, waterfalls (e.g., Seven Sisters).
- **Mawlynnong** (East Khasi Hills) is the 'cleanest village in Asia' with bamboo dustbins; **living root bridges** (near Nongriat, Cherrapunji) are made by weaving tree roots over many years; the Khasi festival **Shad Suk Mynsiem** thanks nature.

1.9 Why Diversity Matters

- Diverse landforms create different soils, plants, animals, jobs and cultures; food habits and food preservation adapt to local conditions (e.g., drying vegetables for the off-season).
- India is called a **'mini-continent'** because it holds mountains, plains, desert, plateau, coasts and islands in one country.
- Geography has shaped civilisation; rivers and shared land help link people into one nation.

1.10 Quick Revision

Likely question	Model answer
Why are the Himalayas called the 'Water Tower of Asia'?	Their snow-melt feeds the Ganga, Indus and Brahmaputra, which supply water to hundreds of millions of people.
How were the Himalayas formed?	The Indian landmass broke from Gondwana, moved north and collided with Eurasia about 50 million years ago; the land between folded upward.
Name the three ranges of the Himalayas.	Himadri (Greater), Himachal (Lower) and Shivalik (Outer).
Why is Ladakh called a cold desert and 'moonland'?	It has very low rainfall and winter temperatures below -30 °C, and its eroded rocky terrain looks like the moon's surface.

Likely question	Model answer
Why are the Gangetic plains densely populated?	Rivers deposit fertile silt and give water; flat land supports farming, roads, railways and trade.
How do people of the Thar adapt to water scarcity?	They use sand to clean utensils, reuse rinse water and store rain in taankas or kunds.
What role do the Aravallis play?	They check the eastward spread of the Thar, hold minerals like marble, zinc and copper, and shelter forts such as Kumbhalgarh.
Why is the southern plateau called the Peninsular Plateau?	It is surrounded by water on three sides: the Arabian Sea, Bay of Bengal and Indian Ocean.
Differentiate between east-flowing and west-flowing peninsular rivers.	East-flowing rivers (Godavari, Krishna, Kaveri, Mahanadi) drain into the Bay of Bengal; west-flowing rivers (Narmada, Tapti) drain into the Arabian Sea.
Compare the west and east coasts.	The west coast is narrow with estuaries and ports; the east coast is wider with large deltas and lakes like Chilika and Pulicat.
Name the two island groups of India with one feature each.	Lakshadweep: coral islands in the Arabian Sea; Andaman and Nicobar: volcanic islands in the Bay of Bengal with Barren Island, India's only active volcano.

★ KEY TAKEAWAYS

- India is the **7th-largest country** and part of the **Indian Subcontinent**.
- Five zones: **mountains, plains, desert, peninsular plateau, islands** (plus coasts).
- Himalayas formed by the **India-Eurasia collision ~50 million years ago**; three ranges: **Himadri, Himachal, Shivalik**.
- The **Gangetic plains** are fertile and populous; the **Thar** is a hot desert; the **Aravallis** are among the oldest hills.
- The **Peninsular Plateau** is flanked by the **Western and Eastern Ghats**; Godavari, Krishna, Kaveri flow east; Narmada and Tapti flow west.
- Coastline **7,500+ km**; west coast has estuaries, east coast has deltas.
- Islands: **Lakshadweep (coral, 36)** and **Andaman and Nicobar (volcanic, 500+)**; Barren Island is the only active volcano.
- Diversity gives India its mix of soils, livelihoods and cultures: a **'mini-continent'**.

CHAPTER 02 · GEOGRAPHY

Understanding the Weather

Weather decides what we wear, eat, grow and plan. This chapter explains what weather is, the five elements that make it up, the instruments used to measure each one, and why accurate forecasting protects lives and livelihoods.

Big Questions

- How can we measure and monitor the weather around us?
- How do weather predictions help us prepare for heavy rain, storms, drought and heat waves?

2.1 Weather and Its Elements

- **Weather** is the state of the Earth's atmosphere at a particular time and place.
- The **atmosphere** is the layer of gases (air) around the Earth, like a cake with layers. The lowest layer is the **troposphere**, where all land life exists and almost all weather happens.
- The troposphere reaches **6 to 18 km** above the ground: thinner at the poles (cold air contracts) and thicker in the tropics (warm air expands). Above it lie the stratosphere (with the ozone layer), separated by the tropopause.

Element	Meaning
Temperature	How hot or cold the atmosphere is
Precipitation	Any water falling from the sky: rain, snow, sleet (frozen or partly frozen rain), hail (small hard balls of ice)
Atmospheric pressure	Weight of the air above us, felt on the Earth's surface
Wind	Movement of air: speed and direction
Humidity	Amount of water vapour (water in gaseous form) in the air

REMEMBER

Words like 'chilly' or 'pleasant' are personal. Weather must be measured with **common standards** so that Krishnan in Chennai and Amir in Kashmir mean the same thing by '20 °C'.

2.2 Traditional Forecasting and Meteorology

- People have long **forecast** (predict in advance) weather by watching nature: birds flying low, **ants moving eggs to higher ground**, frogs croaking loudly, **pine cones** closing in humid air (to protect seeds) and opening in dry air. Such knowledge passes down generations and is still used, especially to guess the monsoon's arrival.
- **Meteorology** is the systematic study of weather and its evolution; it is the basis of forecasting. Meteorologists use instruments to predict weather hours, days or weeks ahead.

2.3 Weather Instruments

a) Temperature: thermometer

- Thermometers (clinical, laboratory, digital) measure temperature; many use a coloured liquid that **expands on heating**. Digital ones are more precise and store more data.
- Scales: **Celsius (°C)** and **Fahrenheit (°F)**; 15 °C = 59 °F.
- Some record **ambient** (immediate surroundings) temperature, others the maximum and minimum of a day.

Statistic	How to find it	Example (max 30, min 15)
Range of temperature	Maximum - minimum in 24 hours	30 - 15 = 15 °C
Mean daily temperature	(Maximum + minimum) / 2	(30 + 15) / 2 = 22.5 °C

> **NOTE**
> The **India Meteorological Department (IMD)** was set up in **1875**. Its motto, from the *Manusmriti*, is *adityat jayate vrishti* — 'from the sun arises rain'.

b) Precipitation: rain gauge

- A **rain gauge** has a funnel that collects rain into a measuring cylinder with a scale; the **depth** of water is the rainfall. 5 mm of water = 5 mm of rainfall; '30 mm in a day' is measured the same way.
- Place it in **open ground**, on a **flat, firm surface**, away from trees and buildings (an elevated platform in open ground is best). Snow must be melted before measuring.

c) Atmospheric pressure: barometer

- Pressure is the weight of air above and around us. It is **higher near the sea coast and lower at height**, because the air gets thinner in the mountains.
- At altitude there is less oxygen, so people feel breathless, dizzy and tired; they must **acclimatise** (adjust) by pausing on the way. Soldiers at **Khardung La** (over 5,600 m, Ladakh) live where pressure is only about **650 mb**.
- Measured by a **barometer** in **millibars (mb)**. Normal sea-level pressure is about **1013 mb**; below **1000 mb** indicates a **depression / low-pressure system**, which can grow into a storm or **cyclone**.

d) Wind: wind vane and anemometer

- **Wind** is air moving from **high pressure to low pressure** areas; described by speed and direction.
- It matters for forecasting, pilots, sailors, seed dispersal, and farmers (wind direction hints where rain will come from; strong wind dries soil faster).
- **Wind vane**: a rotating arm whose pointer turns to show direction. A **wind sock** at airports guides pilots during take-off and landing and is also used where ash or gases are released.
- **Anemometer**: three or four cups on a vertical shaft spin faster in stronger wind; a meter counts rotations to give speed in **km/h**.

e) Humidity: hygrometer

- Humidity is the water vapour in air; it depends on temperature, wind, pressure and location. Measured as **relative humidity**: 0% = no vapour (impossible naturally), **100% = saturated**.
- Dry weather: about **20-40%**; humid weather: about **60-80%**. High humidity means evaporation is slow, so wet clothes dry slowly and sweat does not evaporate easily (you feel sticky). Kochi (84%) is more humid than Delhi (52%).

- Measured by a **hygrometer**. Industries (food processing) and museums (keeping exhibits dry) monitor it.

Instrument	Measures	Unit / note
Thermometer	Temperature	°C or °F
Rain gauge	Precipitation (rainfall)	mm
Barometer	Atmospheric pressure	millibar (mb)
Wind vane / wind sock	Wind direction	—
Anemometer	Wind speed (and direction)	km/h
Hygrometer	Humidity	% relative humidity

2.4 Weather Stations

- A **weather station** brings all instruments together and records readings at regular intervals to help map and forecast weather.
- An **Automated Weather Station (AWS)** uses sensors to record temperature, humidity, wind, precipitation and pressure **without human intervention**; used in agriculture, aviation, navigation and environment monitoring.
- In **2023** the National Disaster Management Authority set up an AWS at a glacial lake in **Sikkim** (above 4,800 m) to give early warning.

2.5 Predicting the Weather

- Meteorologists study long-term data scientifically. Forecasts matter more now because **climate change** is making droughts, floods and cyclones more frequent.
- Uses: governments mobilise resources and plan for disasters; **fishermen are warned** not to go to sea in storms; coastal areas are **evacuated** before a cyclone; farmers plan sowing and harvest; flights and ships plan routes.
- The IMD issues colour-coded warnings by region: **Watch (be updated), Alert (be prepared), Warning (take action)** for heat waves, cold waves, heavy rain, fog, hailstorm, dust storm, thunder and lightning etc. (e.g., the map of 19 May 2024).

IMPORTANT

Together temperature, humidity, precipitation, wind and pressure define weather. At different times a different element dominates: rainfall in July, temperature in May/December, pressure when a cyclone moves, wind during a **loo** (strong, hot, dusty winds of north India in summer). Weather is closely linked to **climate** (next chapter).

2.6 Quick Revision

Likely question	Model answer
Define weather and atmosphere.	Weather is the state of the atmosphere at a given time and place; the atmosphere is the layer of gases (air) surrounding the Earth.
Which layer of the atmosphere has weather?	The troposphere, 6 to 18 km thick.

Likely question	Model answer
List the five elements of weather.	Temperature, precipitation, atmospheric pressure, wind and humidity.
Match instruments with elements.	Thermometer - temperature; rain gauge - precipitation; barometer - pressure; anemometer - wind speed; wind vane - wind direction; hygrometer - humidity.
How do you calculate the range and mean daily temperature?	Range = maximum - minimum; mean = (maximum + minimum) / 2.
Why do mountaineers feel breathless at height?	Air pressure is lower and the air thinner, so less oxygen reaches the blood.
What pressure indicates a depression?	Below 1000 mb (normal at sea level is about 1013 mb).
What is wind?	Movement of air from high to low pressure areas.
Where should a rain gauge be placed?	On open ground, on a flat firm surface or elevated platform, away from trees and buildings.
What does 84% humidity at 29 °C in Mumbai in June suggest?	Very humid, sticky weather: light cotton clothes and rain gear are advised.
Why are weather forecasts useful?	They warn of storms, cyclones, floods and heat waves so that people, fishermen, farmers and governments can prepare and save lives and property.

★ KEY TAKEAWAYS

- Weather = state of the atmosphere at a time and place; happens in the **troposphere**.
- Five elements: **temperature, precipitation, pressure, wind, humidity**, each with its own instrument.
- Thermometer, rain gauge, barometer (mb), anemometer/wind vane, hygrometer.
- Normal pressure ~**1013 mb**; below 1000 mb = depression; pressure falls with height.
- **AWS** records data automatically; the **IMD (1875)** issues warnings.
- Forecasting is vital in an age of climate change; traditional signs complement science.

CHAPTER 03 · GEOGRAPHY

Climates of India

Weather changes by the hour, but climate is the long-term pattern of a place. India's mountains, desert, plateau and long coast give it many climates, and the monsoon ties them together. This chapter explains the factors that shape climate, how the monsoon works, how climate affects life, how climate-linked disasters arise, and what climate change means.

Big Questions

- What makes India's climate so diverse?
- What are the monsoons and how are they formed?
- What is the effect of climate on economy, culture and society?
- How can understanding climate help us prepare for natural disasters?
- What is climate change and what are its consequences?

3.1 Weather, Seasons and Climate

Term	Meaning	Time scale
Weather	Day-to-day condition: rain, sunshine, wind etc.; keeps changing	Hours / days
Season	Part of the year with a typical weather, recurring yearly as the Earth **revolves around the Sun**	A few months, every year
Climate	Pattern of weather of a region over a **long period (three decades or more)**	Decades

- Most of the world has four seasons: spring, summer, autumn, winter. India adds the **rainy season (monsoon)**.
- Traditional Indian calendar: **six ritus** — *vasanta* (spring), *grishma* (summer), *varsha* (rainy), *sharad* (autumn), *hemanta* (pre-winter), *shishir* (winter). Festivals like **Vasanta Panchami** and **Sharad Purnima** go with them.
- Crops, food, clothes, tree-blooming, leaf-fall and animals' thick fur all follow the seasons.
- Climate is usually stable, but in recent decades it has been changing, largely due to human action.

3.2 Types of Climate in India

Region	Climate	Features / link
High Himalayas	Alpine (name from the Alps)	Cold snowy winters, cool summers; thickest clothing
Lower Himalayas and hilly areas	Temperate	Moderately cold winters, not-too-hot summers; hill stations
Northern plains	Subtropical	Very hot summers, cold winters; most of India's **wheat**
Thar Desert (west)	Arid	Extremely hot days, cool nights, very little rain; water-harvesting methods

Region	Climate	Features / link
Western coastal strip	Tropical wet	Heavy monsoon rain; **rice and spices**
Deccan Plateau (central)	Semi-arid	Hot summers, mild winters, moderate rain
Eastern India and southern peninsula	Tropical	Mild winter; distinct wet and dry periods controlled by monsoon winds

3.3 Factors Determining Climate

▸ a) Latitude

- Latitude = distance from the **Equator**. Low latitudes are hot, middle temperate, high (poles) cold.
- Reason: at the Equator the Sun's rays fall nearly **perpendicular** and are concentrated on a small area; near the poles they are **oblique**, spread over a larger area and travel through more atmosphere, losing energy.
- Example: Kanniyakumari and the Nicobar Islands (near the Equator) are warm all year, while Srinagar is much cooler.

▸ b) Altitude

- Temperature **falls with height**. Reasons: air pressure and density decrease with height and thinner air is cooler; and the Sun heats the Earth's surface first, so air farther from it is cooler.
- Hill stations: Munnar, Ooty (Udhagamandalam), Madikeri, Mahabaleshwar, Mount Abu, Shimla, Nainital, Darjeeling, Tawang, Shillong. High Himalayan peaks stay below freezing, hence snow.
- Ooty and Coimbatore are at nearly the same latitude, yet Ooty's summer is 10-25 °C and Coimbatore's 25-38 °C because Ooty is higher.

▸ c) Proximity to the sea

- The sea is a **moderator**: water heats and cools **slowly**, land heats and cools **quickly**. So coasts have mild summers and winters (small range); inland places have extremes.

City	Position	Summer	Winter	Range
Mumbai	Coastal	about 32 °C	about 18 °C	about 14 °C
Nagpur	Inland (similar latitude)	up to 44 °C	about 10 °C	about 34 °C

▸ d) Winds

- Winds carry warm or cool air and moisture. Westerly winds from the deserts of **Arabia and Afghanistan** bring dry hot air to Punjab, Haryana, Rajasthan and Madhya Pradesh (**heat waves**). In winter, cold winds from beyond the Himalayas cause **cold waves** in the foothills.
- Sea winds bring moisture and rain; **monsoon winds** are the best example.

▸ e) Topography

- **Topography** = physical surface features (mountains, valleys, coasts). The **Himalayas and Karakoram** shield India from cold winds of Central Asia; the flat **Thar** has no such shield; the **Western Ghats** shape the monsoon.

DEFINITION

Microclimate: climate of a small area that differs from its surroundings, e.g. enclosed valleys, forests and **urban heat islands** (cities with dense concrete and little greenery trap heat and are warmer than the surroundings). Microclimates affect local plants, animals, crops and health.

3.4 The Monsoons

- 'Monsoon' comes from the Arabic **mausim** (season): seasonal winds over the Indian Ocean region (Australia, Africa, South Asia). Popularly the word means the **rains**.
- Mechanism: land heats and cools faster than ocean.

	Summer (Southwest) monsoon	Winter (Northeast) monsoon
Land vs ocean	Land very hot: **low pressure**; ocean cooler: high pressure	Land cools quickly: **high pressure**; ocean warmer: lower pressure
Wind direction	Ocean to land (from the southwest)	Land to ocean (from the northeast)
Result	Moist winds condense into **heavy rain**	Dry, cold winds; part crossing the **Bay of Bengal** gathers moisture and rains on **east and south India**
Timing	Reaches southern tip in **early June**, covers the subcontinent by **mid-July**	Winter months

- The advance is uneven: the **Western Ghats** block the winds, so their western slopes get heavy rain, while the Deccan plateau (rain-shadow side) gets less and with breaks.
- **Mawsynram** (Meghalaya) has the highest average annual rainfall in the world: about **11,000 mm**.
- Monsoons inspired ragas such as *Meghamalhar* and *Amruthavarshini*.
- Traditional signs: Konkan fishermen see deep-water fish near the surface; in south India the monsoon is said to come within about 50 days of the **Golden Shower (Cassia fistula)** blossoming; crows nesting high suggests little rain, low nests heavy rain. This heritage should be preserved.

3.5 Climate and Our Lives

- **Culture**: festivals follow seasons and farming — Baisakhi, Bihu, Pongal, Makar Sankranti, Onam, Lohri, Gudi Padwa, Chhath Puja, Hemis, Losoong etc.
- **Economy**: a **monsoon failure** (poor rain) hurts agriculture, forces women to walk far for water, pushes labourers to migrate to cities, and raises food prices (**inflation**). Industry also needs steady weather and water.

3.6 Climate and Disasters

Disaster	What it is and causes	Where / impact
Cyclone	Intense low-pressure system over the sea; moist air rushes in and spirals inward. Calm centre is the **eye**. IMD tracks formation, path and landfall	Eastern coast mainly; loss of life, damaged property, uprooted trees, erosion

Disaster	What it is and causes	Where / impact
Flood	Water overflows onto dry land: heavy rain run-off, rivers/lakes breaching banks. In cities: poor drainage, encroachment, concrete surfaces that stop absorption. In Himalaya: **glacial burst** when a glacial lake breaks its rock-ice barrier	UP, Bihar, Kerala, Andhra Pradesh, Assam. **2013 Uttarakhand**: glacial burst, Kedarnath area destroyed, about 6,000 lives lost
Landslide	Sudden collapse of rock, soil or debris after heavy rain, earthquakes or volcanic activity; worsened by **deforestation**, unplanned construction and blocked water flow	Himachal, Uttarakhand, Sikkim, Arunachal, Western Ghats; mostly in monsoon
Forest fire	Uncontrolled fire fuelled by dry weather, drought, high winds and human carelessness	Uttarakhand, Himachal, MP, Chhattisgarh, Western Ghats; kills wildlife, harms air quality, displaces people

- **Clouds** are masses of water droplets, ice crystals or both, suspended in the atmosphere.
- The **National Disaster Response Force (NDRF)**, trained for natural and human-made disasters, has battalions at 12 locations and does rescue and evacuation.

3.7 Climate Change

- **Climate change** = significant long-term shifts in temperature, precipitation and weather events. Earlier driven by natural processes; since the **19th century** largely by humans: burning **fossil fuels** (coal, oil, gas formed from ancient plants and animals), deforestation, harmful industry, wasteful consumption.
- The natural **greenhouse effect** (CO_2 and other gases trapping the Sun's heat) keeps Earth warm enough for life. Human emissions have added huge quantities in a few centuries, trapping extra heat: **global warming**.
- In early 2025 India's average temperature was 1-3 °C above normal, giving a short, mild winter that hurt farming and small industries.
- Responses: **mitigation** (cut emissions, plant trees, renewable energy, energy efficiency, sustainable lifestyles) and **resilience / adaptation** (capacity to withstand and recover). These often clash with the urge for economic growth and consumption.

3.8 Quick Revision

Likely question	Model answer
Differentiate weather and climate.	Weather is the day-to-day state of the atmosphere; climate is the long-term (30+ years) pattern of weather of a region.
Name the six traditional ritus.	Vasanta, Grishma, Varsha, Sharad, Hemanta and Shishir.
List the factors that determine climate.	Latitude, altitude, proximity to the sea, winds and topography.
Why is Chennai hot throughout the year while Leh is cold?	Chennai is near the Equator (low latitude) and on the coast; Leh is at high altitude and farther north.
Why do coastal areas have milder temperatures?	The sea heats and cools slowly and moderates the temperature, so the range is small (Mumbai 14 °C vs Nagpur 34 °C).
How is the southwest monsoon formed?	In summer the Asian land heats up, creating low pressure; moist winds from the cooler ocean rush in and give heavy rain.

Likely question	Model answer
What is the northeast (winter) monsoon?	In winter, winds blow from land to sea; those crossing the Bay of Bengal pick up moisture and rain on east and south India.
What role do the Western Ghats play?	They block monsoon winds, giving heavy rain on the western slopes and less rain on the Deccan side.
What is a microclimate? Give an example.	A local climate different from its surroundings, e.g. an urban heat island.
What are the effects of monsoon failure?	Poor crops, water scarcity, migration of labourers and higher food prices (inflation).
Explain how a cyclone forms.	A deep low-pressure system forms over the sea; moist air flows in and spirals around the cloud-free eye.
What is a glacial burst? Give an example.	A glacial lake breaks its rock-and-ice barrier and floods downstream, e.g. Uttarakhand 2013.
What causes climate change and what is mitigation?	Burning of fossil fuels, deforestation etc. increase greenhouse gases; mitigation means steps to reduce emissions, like renewable energy and planting trees.

★ KEY TAKEAWAYS

- Weather is short-term, seasons recur yearly, **climate** is the long-term pattern (30+ years).
- Five factors: **latitude, altitude, nearness to sea, winds, topography**.
- Seven climate types: alpine, temperate, subtropical, arid, tropical wet, semi-arid, tropical.
- **Monsoon** (Arabic *mausim*): southwest (June-July, rain) and northeast (winter, rain in east/south); Mawsynram receives ~11,000 mm.
- Climate shapes festivals, farming, food prices and the economy.
- Climate-linked disasters: **cyclones, floods, landslides, forest fires**; human action worsens them.
- Climate change is driven mainly by fossil fuels; needs **mitigation and resilience**.

CHAPTER 04 · CULTURE & GEOGRAPHY

How the Land Becomes Sacred

In India, faith is tied to the land itself: to mountains, rivers, trees and forests as well as to shrines. For at least 3,000 years pilgrims have criss-crossed the Subcontinent, and in doing so they turned the whole geography into a sacred landscape and helped bind its people into one culture. This chapter explains how.

Big Questions

- What is 'sacredness'?
- How does the land become sacred?
- How do sacred sites and pilgrimage networks connect with people's life and culture?
- What role did sacred geography play in the cultural integration of the Indian Subcontinent?

4.1 What Is Sacredness?

- **Sacredness** = something of deep religious or spiritual importance, worthy of reverence. It may be a **shrine**, a **journey (pilgrimage)**, the **route**, or even the **land** itself.
- A **pilgrimage** is a journey to a sacred place significant to a religion or belief. A **shrine** is a place regarded as holy through association with the divine, a relic or a spiritual figure. A **relic** is a part of a saint's body or belongings kept for veneration.

4.2 Sacred Places of Different Faiths

Faith	Sacred places / features
Religions from outside India (Islam, Christianity, Judaism, Zoroastrianism)	Visited by people of all faiths, e.g. **Dargah Sharif, Ajmer** (Rajasthan) and **Velankanni Church** (Tamil Nadu)
Buddhism	Places visited by the Buddha or holding relics: **Great Stupa, Sanchi** (relic stupa, MP); **Mahabodhi Stupa, Bodh Gaya** (Bihar), where the Buddha attained enlightenment; over **4 million** visitors a year
Sikhism	**Takhts** (seats of spiritual authority): **Takht Sri Patna Sahib**, **Akal Takht** (Golden Temple, Amritsar), **Takht Sri Keshgarh Sahib** (Anandpur). Guru Nanak visited Haridwar, Prayag, Mathura, Varanasi, Ayodhya, Puri and some Muslim shrines
Jainism	Tirthas where **Tirthankaras** attained liberation or key events occurred: **Mount Abu, Girnar, Shatrunjaya** (Saurashtra, Gujarat); trees, ponds, hills they visited are sacred
Hinduism, folk and tribal beliefs	Mountains, rivers, forests, trees, animals and stones; the Earth as **Mother Earth / Bhudevi**. Example: Sabarimala (Ayyappa), over 10 million devotees yearly, a difficult hill trek symbolising the inner path

DEFINITION

Tirtha: literally a place to *cross* a river or water; symbolically a place where one crosses from ordinary worldly life to a higher spiritual one. **Tirthayatra** = pilgrimage to tirthas; it is an outer journey and an inner one needing a code of conduct. **Tirthankara** (Jainism): one who makes a tirtha, guiding the crossing to a higher life; supreme preacher of dharma.

- Pilgrimage spots range from **Badrinath, Kedarnath, Amarnath** in the Himalayas to **Kanyakumari**. Jawaharlal Nehru (1961) said the feeling of one country and one culture drew people north-south on pilgrimages.
- **Pandharpur wari** (Maharashtra): an 800-year-old yearly walk of 21 days to the Vithoba temple. *Wari* = a regularly held pilgrimage.
- Dharampal's account: villagers from north of Lucknow on a three-month pilgrimage to Rameswaram, carrying atta, ghee, sugar and earthen pots, changing trains at Delhi to go straight to Haridwar — showing how ordinary people still travel long distances for faith.

4.3 Nature as Sacred

- Hindu, folk and tribal traditions see divinity in **all Nature**; rivers are *devis*, certain trees and animals are holy; Earth is **Bhudevi** (shown in Vishnu's boar avatar rescuing her, Belur temple).
- **Niyam Dongar** hill (Niyamgiri, Odisha): abode of Niyam Raja for the **Dongria Khond**; cutting trees is disrespect.
- **Sikkim** (early 2000s) identified sacred mountains, caves, lakes, rocks and hot springs to protect.
- **Todas** of the Nilgiris regard peaks, plants, Shola forests, wetlands, stones and trees as sacred.

4.4 Sacred Geography

- Pilgrimage networks cover India end to end and make the **land itself sacred**: the **Char Dham** (four sites placed in the four corners: north, south, east, west), the **12 Jyotirlingas** (shrines of Shiva) and the **51 Shakti Pithas** (even in present Bangladesh and Pakistan), plus many regional circuits.
- **Shakti Pitha story**: Sati, insulted by her father, immolated herself; a grief-stricken Shiva carried her body; Vishnu cut it with his chakra and the parts fell at 51 places. Symbol: the whole land is the body of the divine Mother.
- Pilgrims met different languages, customs, clothes and foods yet noticed common threads. Traders travelled to exchange goods; scholars to debate, teach and learn; routes converged and ideas mixed: a major cause of **cultural integration** of the Subcontinent.

4.5 Sacred Ecology

- Tirthas lie on **river or lake banks, in forests and on mountains**; such land is a **punyakshetra**. Seeing Nature as sacred helped to preserve it.
- **Rivers**: worshipped since Vedic times; the *Nadistuti Sukta* of the **Rigveda** praises **19 rivers** of the northwest. A prayer invokes Ganga, Yamuna, Godavari, Sarasvati, Narmada, Sindhu and Kaveri in the bathing water. Rivers are addressed respectfully ('Ganga ji'). **Sangams** (confluences) are holy.
- **Kumbh Mela**: held in rotation at **Haridwar, Prayagraj, Nashik, Ujjain** (Prayagraj every six years, at the Ganga-Yamuna-invisible Sarasvati confluence); UNESCO lists it as intangible heritage; about **660 million** took part in 2025. Legend: devas and asuras churned the cosmic ocean (*amrita manthana*); Vishnu as **Mohini** seized the pitcher (*kumbha*) of nectar and drops fell at these four places.
- **Mountains**: seen as a gateway from earth to heaven; hilltop shrines mean the climb is a symbolic journey (Mount Kailash, Vaishno Devi, Tirumala, Tiruvannamalai).

- **Trees**: the **peepul** (*Ficus religiosa*, *ashvattha*, bodhi tree) is sacred to Hindus, Buddhists, Sikhs and Jains; the Mahabodhi tree at Bodh Gaya is a descendant of the Buddha's tree; peepul leaves appear on a **Mohenjo-daro seal**. It has medicinal uses and shelters birds.
- **Epics**: Mahabharata and Ramayana describe pilgrimages; local shrines (e.g. Bastar's Rama shrine) let many communities make the epics their own.

▸ Sacred groves

- Forests protected by communities as abodes of deities (e.g. *Ryngkew / Basa* in Meghalaya): no hunting, tree felling or mining. They shelter great biodiversity and small water bodies (water conservation). Thousands once existed but are shrinking due to encroachment.
- Thanjavur chronicles: grove deities protect **fruit bats**, which pollinate flowers and spread seeds.

Language / region	Name for sacred grove
Malayalam	kavu
Tamil	kovilkadu
Kannada	devara kadu
Marathi	devrai
Khasi (Meghalaya)	khlaw kyntang
Himachal (Hindi)	dev van
Jharkhand	sarna
Chhattisgarh	devgudi
Rajasthan	oran

4.6 From Pilgrimage to Trade

- Pilgrims need goods and traders supply them; pilgrim routes and **trade routes overlap**, and traders often double as pilgrims.
- Routes: **Uttarapatha** (north-west to east) and **Dakshinapatha** (Kaushambi through Ujjayini/Ujjain to Pratishthana/Paithan). Goods: shells, pearls, gold, diamonds, coins, cotton, spices, sandalwood. Sacred sites therefore boosted local economies.

4.7 Beyond India and Today

- Sacred geography exists elsewhere: ancient Greece (sacred mountains, groves), Native Americans, and the **Maoris** of New Zealand, who see **Taranaki Maunga** as an ancestor; a law has granted the mountain the rights of a human being, with community elders speaking for it.
- Rivers such as the Yamuna, Mahanadi and Kaveri are polluted today; groves are encroached. Conserving sacred places is a shared duty (the Constitution also asks us to protect heritage). David Suzuki: how we see the world shapes how we treat it — a mountain as a deity, a river as a vein of the land. Sacred geography can support **sustainability**.

4.8 Quick Revision

Likely question	Model answer
What is sacredness?	Deep religious or spiritual significance worthy of reverence — a place, journey, route or land itself.
Define pilgrimage, shrine and relic.	Pilgrimage: journey to a sacred place; shrine: holy place linked to divine/relic/spiritual figure; relic: part of a saint's body or belongings kept for reverence.
What does tirtha mean?	A crossing place; symbolically where one crosses from worldly life to a higher spiritual life.
Name Buddhist, Sikh and Jain sacred sites.	Sanchi stupa and Bodh Gaya; Akal Takht, Patna Sahib, Keshgarh Sahib; Mount Abu, Girnar, Shatrunjaya.
How do Char Dham and Shakti Pithas create sacred geography?	They are located across the length and breadth of India, so pilgrims travel the whole land, making it sacred.
Narrate the story of the Shakti Pithas.	Sati immolated herself after her father's insult; Shiva carried her body; Vishnu cut it with his chakra and the parts fell at 51 places.
What is the Kumbh Mela and where is it held?	A great festival based on the amrita legend, held at Haridwar, Prayagraj, Nashik and Ujjain; UNESCO lists it as intangible heritage.
Why is the peepul tree sacred?	It is holy to Hinduism, Buddhism, Sikhism and Jainism; the Buddha attained enlightenment under a bodhi tree; it is useful and shelters wildlife.
What are sacred groves and why matter?	Forests protected as abodes of deities; they conserve biodiversity and water.
How did pilgrimage help trade and integration?	Pilgrim and trade routes overlapped; travellers exchanged goods, ideas and customs, fostering cultural unity.
What is the two-fold significance of tirthayatra?	Individual spiritual growth and socio-economic/cultural benefits such as trade and unity.
How does the Maori example resemble Indian beliefs?	Taranaki Maunga is treated as a living ancestor with legal rights, like Indian traditions that regard mountains and rivers as sacred.

★ KEY TAKEAWAYS

- All faiths in India have sacred places; Buddhist, Jain and Sikh sites link to great figures.
- **Tirtha** = crossing place; **tirthayatra** is an outer and inner journey.
- **Char Dham, 12 Jyotirlingas, 51 Shakti Pithas** make the whole land a sacred geography.
- Rivers, sangams, mountains, peepul and **sacred groves** protect Nature.
- **Kumbh Mela** at Haridwar, Prayagraj, Nashik, Ujjain.
- Pilgrim and **trade routes** overlapped, driving **cultural integration**.
- Today, sacred rivers and groves need protection; sacred geography supports sustainability.

CHAPTER 05 · HISTORY

New Beginnings: Cities and States

After the Harappan cities faded, India had no urban life for about a thousand years. From the 1st millennium BCE, cities rose again in the Ganga plains, along with organised states called janapadas and mahajanapadas. This 'Second Urbanisation' brought iron tools, coins, trade routes, new ideas and one of the world's earliest republics.

Big Questions

- What is meant by the 'Second Urbanisation of India'?
- Why were janapadas and mahajanapadas important in early Indian history?
- What system of governance did they evolve?

5.1 From First to Second Urbanisation

- The Indus / Harappan / Sindhu-Sarasvati civilisation (**First Urbanisation**) declined in the early 2nd millennium BCE: planned cities, markets, specialised crafts, writing, sanitation, administration and the ruling class vanished; people reverted to village life.
- For nearly a **thousand years** urban life was largely absent (a few towns only).
- In the **1st millennium BCE** a new urban phase began in the **Ganga plains**, parts of the Indus basin and nearby regions, then spread. It is the **Second Urbanisation**, which, in a way, continues today.
- **Sources**: (1) **archaeological excavations** of ancient urban centres; (2) **literature** — late Vedic, Buddhist and Jain texts mention them.

5.2 Janapadas and Mahajanapadas

- Towards the end of the 2nd millennium BCE clans (sharing language and customs) became linked to territories called **janapadas**, ruled by a **raja**. *Janapada* = 'where the people (*jana*) have set foot (*pada*)', i.e. settled.
- Trade networks helped them grow; by the **8th-7th centuries BCE** some merged into larger states: **mahajanapadas**. The usual list has **16**, from **Gandhara** (north-west) to **Anga** (east) and **Ashmaka** (central India near the Godavari).
- Many lay in the **Ganga plains** because of fertile land (farming), **iron ore** in nearby hills, and new trade networks.
- Most powerful: **Magadha, Kosala, Vatsa, Avanti**.

Mahajanapada	Capital (and note)	Region
Magadha	Rajagriha (modern Rajgir)	Part of Bihar; most powerful
Kosala	Shravasti	Part of Uttar Pradesh
Vatsa	Kaushambi (near Prayagraj)	Uttar Pradesh
Avanti	Ujjayini (Ujjain)	Part of Madhya Pradesh
Gandhara	Takshashila	North-west

Mahajanapada	Capital (and note)	Region
Vrijji (Vajji)	Vaishali	Early republic (gana / sangha)
Malla	Kushinara	Neighbour of Vrijji; early republic
Anga	Champa	East
Others	Kuru (Indraprastha), Panchala (Ahichchhatra), Shurasena (Mathura), Matsya (Viratnagar), Chedi, Kamboja, Ashmaka	—

- Capitals were big, **fortified** cities with a **moat** (deep water-filled ditch) outside the ramparts and **narrow gateways** so guards could control the movement of people and goods. Many of those capitals are living cities 2,500 years later.

5.3 Early Democratic Traditions

- Each janapada had an assembly — **sabha** or **samiti** (the words appear in the Vedas), mostly of elders. The raja was expected to **take advice** from them and from ministers; texts even suggest an incompetent king could be removed (though this was not necessarily settled law, as evidence is incomplete).

Feature	Monarchical mahajanapadas	Ganas / Sanghas (Vrijji, Malla)
Authority	Raja is the ultimate authority, supported by ministers and elders	Sabha / samiti holds more power
How decisions are taken	By the king with advice	Discussion and, if needed, **vote**
Ruler	**Hereditary** (usually son of previous king)	**Selected** by the assembly
Examples	Magadha, Kosala, Avanti	Vrijji and Malla: called **'early republics'**, among the earliest in the world

- Duties of a king: collect taxes (revenue), keep law and order, build fortifications, maintain an army to defend or fight neighbours.

5.4 Further Innovations

- **New ideas**: late Vedic, Buddhist and Jain schools spread through scholars, monks, nuns and pilgrims; art was renewed (to blossom later under empires).
- **Iron metallurgy**: Harappans used copper and bronze. Iron extraction was perfected in several regions from the early 2nd millennium BCE and tools became widespread by the late 2nd millennium BCE. Iron tools allowed **farming on a bigger scale**, and lighter, sharper **weapons** (swords, spears, arrows, shields). Evidence of **warfare** between mahajanapadas; wars and alliances later produced kingdoms and empires.
- **Coins**: made necessary by growing trade. The first were of **silver**, punched with symbols — **punch-marked coins**; later copper, gold. Each mahajanapada issued its own, but coins circulated across regions and abroad.

Feature	First Urbanisation (Harappan)	Second Urbanisation
Region	Indus-Sarasvati valleys	Ganga plains (spreading everywhere)

Feature	First Urbanisation (Harappan)	Second Urbanisation
Metal	Copper / bronze	**Iron** (also copper, bronze)
Money	Seals / barter-based trade	**Punch-marked coins**
Political structure	Cities with administration	Janapadas, mahajanapadas, ganas
Writing / literature	Script yet undeciphered	Vedic, Buddhist, Jain literature
Monasteries and warfare	No	Yes (monasteries; warfare between states)

5.5 The Varna-Jati System

- Complex societies develop groups by occupation or class. Ideally they complement each other, but they also create **inequality** (some groups gain more wealth and power).
- Indian society had a two-fold system: **Jati** = community defined by a specific occupation, skills passed down generations; may split into **sub-jatis** with their own customs (marriage, ritual, food). **Varna** = concept from Vedic texts with **four** varnas.

Varna	Role
Brahmins	Preserve and spread knowledge; perform rituals
Kshatriyas	Defend society and land; fight when needed
Vaishyas	Increase wealth through trade, business, agriculture
Shudras	Artisans, craftspeople, workers, servants

- The word **'caste'** comes from the Portuguese *casta* (16th-century Portuguese travellers). Scholars use it for varna, for jati, or for the whole system.
- In the early period people **changed occupations** when needed (e.g., drought sends farmers to cities; some Brahmins took to trade or military work). The system gave **stability** but later became **rigid**, causing inequality and discrimination against lower jatis and groups outside the system; it hardened especially under British rule. It is not the only mechanism of Indian society.

5.6 Developments Elsewhere in India

- Trade, pilgrimage and military routes opened up: **Uttarapatha** (north-west to Ganga plains and east) and **Dakshinapatha** (from **Kaushambi** across the **Vindhyas** to the south); lateral roads joined western and eastern ports.
- **Shishupalgarh** (Sisupalgarh, Bhubaneswar), capital of **Kalinga**: strict square plan, fortifications with narrow gateways, moat, broad streets (excavated from 1948).
- South India: cities appear from about **400 BCE**; kingdoms of the **Cholas, Cheras and Pandyas** (named in early Tamil literature). South was rich in gems, gold, spices and traded with India and overseas; **Kodumanal** (near Erode) had shell and gemstone industries.
- By **300-200 BCE** the whole Subcontinent, including the Northeast, was interlinked, with goods and culture reaching Central and Southeast Asia. About then the mahajanapadas ended, making way for empires.

NOTE

Timeline: Harappan civilisation (about 1900 BCE) → janapadas (about 1200 BCE) → mahajanapadas (about 800 BCE) → Second Urbanisation in full swing → ends about 300 BCE. Kautilya's *Arthashastra* describes the ideal kingdom: fortified capital, fertile land, mines, forests, pastures, rivers not only rain, good roads and a productive economy.

5.7 Quick Revision

Likely question	Model answer
Why did urban life vanish after the Harappan civilisation?	Its structures — administration, specialised crafts, markets, writing, sanitation and the ruling class — disappeared, and people returned to village life.
What is the Second Urbanisation and how do we know about it?	A new phase of city growth in the 1st millennium BCE in the Ganga plains; known from excavations and from Vedic, Buddhist and Jain literature.
What does janapada mean?	'Where the people have set foot', i.e. a territory settled by a clan.
How many mahajanapadas were there? Name the most powerful.	Sixteen; Magadha, Kosala, Vatsa and Avanti were the most powerful.
Why were many mahajanapadas in the Ganga plains?	Fertile land, iron ore in nearby hills and growing trade networks.
Describe the defence of a mahajanapada capital.	A fortified city with rampart walls, a water-filled moat and narrow gateways to control entry and exit.
How did gana-sangha differ from monarchy?	In ganas (Vrijji, Malla) the assembly took decisions by discussion or vote and chose the ruler; in monarchies the king was hereditary and the final authority.
Why was iron important?	Iron tools expanded farming; iron weapons strengthened armies; both supported urban growth.
What were punch-marked coins?	India's earliest silver coins, marked with punched symbols, made necessary by growing trade.
Explain the varna and jati systems.	Varna: four Vedic classes (Brahmin, Kshatriya, Vaishya, Shudra); jati: occupation-based communities with inherited skills and own customs.
Name two major trade routes.	Uttarapatha and Dakshinapatha.
Which three kingdoms rose in the south?	The Cholas, Cheras and Pandyas.

★ KEY TAKEAWAYS

- Harappan decline → ~1,000 years with little urban life → **Second Urbanisation** (1st millennium BCE, Ganga plains).
- **Janapadas → 16 mahajanapadas** (Magadha, Kosala, Vatsa, Avanti most powerful).
- **Sabha / samiti** advised the raja; **Vrijji and Malla** were early republics (ganas).
- **Iron, punch-marked coins**, trade routes and new ideas drove growth.
- **Varna** (four classes) and **jati** (occupation groups) organised society; early flexibility, later rigidity.
- **Uttarapatha / Dakshinapatha** linked regions; Cholas, Cheras, Pandyas rose in the south.

CHAPTER 06 · HISTORY

The Rise of Empires

Between the 6th and 2nd centuries BCE, small kingdoms merged into large empires, led by Magadha. This chapter explains what an empire is, how Magadha, the Nandas and the Mauryas rose, how Alexander's invasion touched the north-west, what Kautilya taught about governance, why Ashoka chose peace, and why empires are fragile.

Big Questions

- What is an empire?
- How did empires rise and shape Indian civilisation?
- What factors helped kingdoms become empires?
- What was life like from the 6th to the 2nd century BCE?

6.1 What Is an Empire?

- 'Empire' comes from Latin **imperium** = supreme power. An empire is a collection of smaller kingdoms or territories under a powerful ruler (or group), often won by war. The smaller territories keep their own rulers but are **tributaries** (vassals) to the **emperor**, who rules from a capital that is an economic and administrative centre.
- A **tributary** pays **tribute** (money, gold, grain, livestock, elephants, goods) as a sign of submission and accepts the emperor's **overlordship**.
- Sanskrit terms for emperor: **samraj** (lord of all), **adhiraja** (overlord), **rajadhiraja** (king of kings).

▸ Features of an empire

1. Emperor exerts **central authority** over tributary kings.
2. Keeps an **army** to control tributaries, expand and guard against attack.
3. Maintains an **administration** (officials who govern territories, collect taxes, keep law and order).
4. Builds **communication networks** (roads, river and sea routes) and infrastructure for trade and welfare.
5. Makes **laws**, issues **currency**, weights and measures, regulates trade.
6. **Controls resources**: mines, forests, farm produce, manpower.
7. Encourages arts, literature, religions, schools of thought and centres of learning.

- **Why become an emperor?** ambition to rule the 'whole world' and be remembered by posterity; control resources to build economic and military strength; wealth.
- **How empires grow**: trained armies conquer neighbours; fortified settlements on borders; war on neighbours; control of rivers and trade networks (resources and tax); the kingdom with stronger military and surplus resources becomes overlord.

6.2 Trade, Trade Routes and Guilds

- An army is costly (food, pay, weapons, horses, elephants, roads), so empires need strong **production and trade**. More trade → more income for producers and more tax for the king.
- Traded goods: textiles, spices, farm produce, gems, handicrafts, animals; many Indian goods went overseas by land and sea. Major routes: **Uttarapatha, Dakshinapatha**; cities such as Pataliputra, Ujjayini, Taxila, Mathura, Kaushambi, Tamralipti, Bharukachchha, Kanchipuram, Madurai, Muchiri.

- **Guilds (shrenis)** were associations of traders, craftsmen, moneylenders or farmers, headed by an elected head and executive officers. They (1) turned competitors into **collaborators** sharing market information and (2) had **autonomy** to frame their own rules; the king did not interfere. They lasted centuries and show India's **self-organising ability** (as do village councils).

6.3 The Rise of Magadha

- In the 6th-4th centuries BCE, **Magadha** (south Bihar and nearby) rose above the 16 mahajanapadas; **Ajatashatru** was an early powerful king. The Buddha and Mahavira lived in his time.
- **Why Magadha?** resource-rich **Ganga plains** with fertile soil, timber forests and elephants; **iron ore** from nearby hills (iron ploughs raised output; lighter, sharper weapons strengthened the army); the **Ganga and Son** rivers aided transport and trade; surplus food let artisans flourish.
- **Mahapadma Nanda** (5th century BCE) founded the **Nanda dynasty**, united many kingdoms, issued coins and kept a large army (per Greek accounts). The last Nanda, **Dhana Nanda**, was rich but oppressive and unpopular, so the empire fell to the Mauryas.
- Sanskrit grammarian **Panini** (5th century BCE) composed the **Ashtadhyayi**, 3,996 concise **sutras** on grammar.

6.4 The Arrival of the Greeks

- North-west India had smaller kingdoms such as that of the **Pauravas** under **Porus**.
- **Alexander** of Macedonia (334-331 BCE) conquered the Persian Empire, spreading Greek influence over three continents. **327-325 BCE**: he entered India, defeated Porus in Punjab (the Battle of the **Hydaspes**) but faced fierce resistance and lost men; his soldiers, tired and homesick, refused to go further towards the Ganga. He retreated, losing many in the harsh coastal/desert route. He died in **Babylon** at 32 (**324-323 BCE**) and his empire was split among his generals and **satraps** (governors of provinces).
- Porus, asked how he wished to be treated, replied '**Like a king**'; Alexander restored him as satrap.
- **Gymnosophists** ('naked philosophers'), Indian sages, answered Alexander's riddles calmly and impressed him — a meeting of Greek and Indian thought.
- Impact: limited political effect but opened **Indo-Greek cultural contacts**.

6.5 The Mighty Mauryas

- About **321 BCE**, **Chandragupta Maurya**, helped by mentor **Kautilya** (Chanakya / Vishnugupta, teacher at **Taxila**), overthrew Dhana Nanda (who insulted Kautilya, who vowed to end the Nandas) and founded the **Maurya Empire** with **Pataliputra** (modern Patna) as capital.
- He defeated the Greek satraps in the north-west; the empire stretched from the northern plains to the Deccan. Greek ambassador **Megasthenes** lived in his court and wrote **Indika** (lost, known from quotations).

▸ Kautilya's Arthashastra

- *Arthashastra* = 'science of governance and economics': defence, economy, administration, justice, town planning, agriculture, welfare, anti-corruption laws.
- **Saptanga** (seven parts of a kingdom) is shown below.
- He stressed law and order, welfare of the **countryside** (source of economic activity), and said: 'In the happiness of his subjects lies the king's happiness' — the people's interests come first.

Part	Meaning
Swami	The king

Part	Meaning
Amatya	Ministers, councillors and officials
Janapada	Territory and its population
Durga	Fortified towns and cities
Kosha	Treasury
Danda	Forces of defence and law and order
Mitra	Allies

6.6 Ashoka: The King Who Chose Peace

- **Ashoka (268-232 BCE)**, Chandragupta's grandson, expanded the empire over almost all the Subcontinent except the far south, including today's Bangladesh, Pakistan and parts of Afghanistan.
- The **Kalinga war** (modern Odisha) brought huge death and destruction; Ashoka gave up violence and adopted the Buddha's **peace and non-violence**. He admitted this in his edicts.
- He sent **emissaries** to Sri Lanka, Thailand, Central Asia and beyond to spread the Buddha's message.
- **Edicts** (official declarations) were engraved on rocks and pillars in **Prakrit** using the **Brahmi** script (mother of Indian scripts) at Girnar, Sarnath, Sanchi, Dhauli, Lumbini, Kandahar and others; Ashoka is called a 'great communicator'. He styled himself **Devanampiya Piyadasi** (Beloved of the Gods; one who regards all with kindness).
- Welfare claims: medical care for people and animals, ban on hunting and cruelty to animals (early conservation), rest houses, wells and shade trees on roads, encouraging all sects to respect each other. An edict tells officers that all men are his children, to be impartial and never imprison or torture without cause; an officer toured every five years.
- **Dharma (dhamma)**: moral law and duty; at a deeper level, living in harmony with the cosmic order (*ritam*): duty, law, truth, order and ethics.
- After Ashoka the empire survived about 50 years, then weak successors let regions break away; around **185 BCE** a new phase began.

6.7 Life in the Mauryan Period

- Cities such as Pataliputra had palaces, public buildings and planned streets; a strong **taxation system** and trade kept the treasury rich.
- **Sohgaura copper plate** (4th-3rd century BCE, Prakrit, Brahmi): one of the earliest administrative records, about a **granary** for famine relief (food security).
- From Megasthenes: most people farmed; **two crops a year**; famines rare; granaries well stocked; farmers were left undisturbed even in wartime. Artisans (blacksmiths, potters, carpenters, jewellers) lived in planned cities with street signs; couriers carried messages; wooden houses up to two storeys; water vessels kept for fire; cotton clothes and thick-soled leather shoes.

▸ Contributions of the Mauryas

Area	Details
Art and architecture	Highly polished stone pillars; **Sarnath lion capital** with four lions (royal power), an elephant, bull, horse and lion on the ring and the **dharmachakra**; **Great Stupa, Sanchi** (originally brick, later stone); stupas, chaityas, viharas; rock-cut caves at Barabar Hills; Dhauli elephant sculpture
Symbols today	Sarnath capital = **national emblem** with motto **Satyameva Jayate** ('truth alone triumphs', *Mundaka Upanishad*); dharmachakra on the national flag
Economy	Strong trade routes, extensive punch-marked coins, tax system
Urban life and administration	Well-designed towns, elaborate administration
Terracotta art	Dancing girl, female deity, horse head, Saptamatrikas

DEFINITION

Stupa: dome-shaped structure holding relics. Its hemispherical dome is the **anda** (the universe); people walk around it clockwise (**pradakshina**) in worship.

6.8 The Fragile Nature of Empires

- Reasons for decline: regions seeking independence when tribute demands rise (long wars, drought); weak successors; the larger an empire, the harder to hold (distant areas split first); economic crises from floods and droughts.
- Paradox: empires bring **political unity**, peace among small states and prosperity, but are built by **war and force**, making them unstable.

Period (BCE)	Event
600-500	Mahajanapadas; rise of Magadha (Ajatashatru)
5th century	Mahapadma Nanda founds the Nanda dynasty; Panini
327-325	Alexander in north-west India
321	Chandragupta Maurya founds the Maurya Empire
268-232	Ashoka's reign; Kalinga war
185	End of the Mauryas

6.9 Quick Revision

Likely question	Model answer
What is an empire? How does it differ from a kingdom?	An empire is a large territory of many kingdoms ruled by an emperor, with tributary kings; a kingdom is a single state under its own king.
What is tribute?	Money, gold, grain, animals or goods paid by a vassal state to the emperor as a sign of loyalty.

Likely question	Model answer
Why did Magadha become powerful?	Fertile Ganga plains, forests, elephants, iron ore, rivers for trade and a strong army.
What are guilds? Why important?	Associations of traders and craftsmen with elected heads and autonomy; they encouraged cooperation and kept trade strong.
Who were the Nandas and why did they fall?	A dynasty founded by Mahapadma Nanda; the last king, Dhana Nanda, was oppressive and unpopular, and was overthrown by Chandragupta.
What was the effect of Alexander's invasion?	Limited political impact; opened Indo-Greek contacts; his soldiers refused to proceed beyond Punjab.
Who was Kautilya? What is the saptanga?	Mentor of Chandragupta and author of the Arthashastra; saptanga lists seven parts of a state: king, ministers, territory, forts, treasury, army and allies.
Who was Megasthenes?	Greek ambassador at Chandragupta's court who wrote Indika.
Why did Ashoka give up war?	The death and destruction of the Kalinga war changed him; he adopted the Buddha's path of non-violence and dharma.
How did Ashoka spread his message?	Through edicts in Prakrit/Brahmi on rocks and pillars and emissaries sent abroad.
What did Ashoka do for welfare?	Medical care, rest houses and wells, shade trees, protection of animals, tolerance of all sects.
Which Mauryan symbols are used today?	The Sarnath lion capital as the national emblem, with Satyameva Jayate; dharmachakra on the flag.
Why are empires fragile?	They rest on force; distant regions rebel; weak rulers, tribute burdens or disasters cause break-up.

★ KEY TAKEAWAYS

- Empire = many kingdoms under one emperor; vassals pay **tribute**.
- **Magadha**: Ganga plains, iron, rivers → Nandas → **Mauryas (321 BCE)**.
- **Guilds** and trade routes funded empires.
- **Alexander (327-325 BCE)** reached Punjab; no lasting rule in India.
- **Kautilya's Arthashastra** and **saptanga** shaped Mauryan governance.
- **Ashoka** (268-232 BCE): Kalinga war → peace, dharma, edicts in Prakrit/Brahmi.
- Legacy: **national emblem**, dharmachakra, Sanchi, roads and coins; empires are strong yet **fragile**.